

Harmain Ali Butt 1781910874

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PROFESSIONAL SUMMARY

Seasoned Lead Software Engineer with 8+ years of expertise in developing, deploying, and optimizing Artificial Intelligence (AI) and Machine Learning (ML) models for high-performance production environments. Proficient in Python, C++, and JavaScript, with a deep understanding of data science principles, linear algebra, algorithms, and statistics. Proven track record of leveraging frameworks like PyTorch and TensorFlow to build scalable ML/AI solutions and creating interactive web applications using Flask and Angular. Committed to ensuring data quality and model reliability through rigorous quality control and automated monitoring processes.

TECHNICAL SKILLS

Python, C++, JavaScript, PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib, Plotly, Flask, Angular, SQL, PostgreSQL, MongoDB, RESTful APIs, Artificial Intelligence, Machine Learning, Data Science, Linear Algebra, Algorithms, Statistics, Quality Control, Model Deployment, Data Analytics Automation, Git, Jupyter Notebooks, VS Code, Google Colab, Docker, AWS

PROFESSIONAL EXPERIENCE

Senior Machine Learning Engineer — AMD (Jan 2024 - Present)

Technologies: Python, PyTorch, TensorFlow, C++, AWS, Docker, SQL, noSQL

Led the development and deployment of advanced Artificial Intelligence (AI) models for optimizing GPU performance, resulting in a 15% improvement in processing efficiency across key product lines. Implemented robust quality control protocols and established automated monitoring systems to meticulously track output data quality in production environments, reducing critical errors by 20%. Architected and developed scalable ML/AI tool development pipelines using Python and C++, integrating complex algorithms and linear algebra concepts to enhance model training and inference capabilities. Collaborated with cross-functional teams to automate data analytics processes, identifying critical opportunities for ML and AI applications that drove strategic product enhancements.

Machine Learning Engineer — NVIDIA (Jan 2021 - Dec 2023)

Technologies: Python, TensorFlow, NumPy, Pandas, JavaScript, Flask, PostgreSQL

Developed and deployed machine learning models for real-time image recognition and natural language processing, significantly improving data processing speeds by 25% using TensorFlow and Python. Contributed to ML/AI tool development, creating custom libraries that integrated advanced statistics and algorithms for feature engineering and model evaluation. Designed and implemented interactive web development interfaces for ML model visualization and interaction using JavaScript and Flask, enhancing user accessibility and data interpretation. Established rigorous quality control measures for deployed models, including continuous monitoring of output data quality in production environments to ensure model integrity and performance.

AI Developer — Google (Jan 2018 - Dec 2020)

Technologies: Python, Scikit-learn, C++, SQL, Angular, ASP.Net Core

Developed core Artificial Intelligence (AI) components for Google Cloud services, focusing on predictive analytics and anomaly detection using Python and C++. Engineered and deployed machine learning models, applying principles of linear algebra and statistics to optimize data-driven decision-making processes. Collaborated on web development projects, building user interfaces with Angular and ASP.Net Core to expose ML model functionalities and insights to internal stakeholders. Participated in the design of data pipelines for automating data analytics, ensuring data quality and consistency for downstream ML applications.

EDUCATION

Master's in Computer Science — Stanford University (2017)

Bachelor of Science in Computer Engineering — University of California, Berkeley (2016)

CERTIFICATIONS

Machine Learning Engineering with TensorFlow on Google Cloud | Google Cloud, Deep Learning Specialization | DeepLearning.AI (Andrew Ng), Professional Certificate in Python for Data Science and AI | IBM, Advanced JavaScript and Front-End Development | Udemy